

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

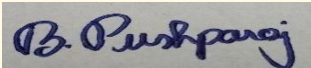
Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1008	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	B. Pushparaj	Reg. No.:	192421136
Date:	28.07.2026	Duration:	60 minutes

Code of Conduct

I B. Pushparaj (192421136)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____  _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

2. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

2. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

2. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

2. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

2. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent(5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task	Backlog prioritization and sprint planning are mostly	Prioritization or sprint planning is partially complete with	Backlog prioritization and sprint planning are poorly organized or incomplete.

	breakdown is complete.	appropriate with minor issues.	limited justification.	
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

1. Project Vision and Stakeholders

Project Vision

Develop an accessible digital learning platform that enables students with visual impairments to access educational resources independently using screen readers, voice navigation, audio learning materials, and accessible online assessments.

Project Objectives

- Provide equal access to education for visually impaired students.
- Support screen readers and keyboard navigation.
- Enable voice-based interaction.
- Offer audio learning resources.
- Conduct accessible online assessments.
- Track student learning progress.
- Improve the overall learning experience.

Stakeholders

Stakeholder	Role
Students	Access courses, audio lessons, quizzes
Teachers	Upload learning materials and quizzes
Administrator	Manage users, courses, reports
Educational Institution	Monitor platform performance
IT Support Team	Maintain the software
Parents	Monitor student progress

2. Epics and User Stories

Epic 1: User Authentication

US1

As a student, I want to register an account so that I can access learning resources.

US2

As a student, I want to log in securely so that I can continue my learning.

Epic 2: Accessible Learning

US3

As a student, I want audio lessons so that I can learn without reading text.

US4

As a student, I want screen reader compatibility so that I can access all content easily.

US5

As a student, I want voice navigation so that I can navigate the platform hands-free.

Epic 3: Online Assessment

US6

As a student, I want accessible quizzes so that I can evaluate my knowledge independently.

US7

As a student, I want to view my quiz scores so that I can monitor my progress.

Epic 4: Administration

US8

As an administrator, I want to manage users so that I can maintain the platform.

US9

As an administrator, I want to upload learning materials so that students always have updated content.

US10

As an administrator, I want to generate reports so that I can monitor student performance.

3. Acceptance Criteria

US1 – Student Registration

Given a new student

When the student enters valid registration details

Then the account should be created successfully.

US2 – Login

Given a registered user

When valid credentials are entered

Then the dashboard should open.

US3 – Audio Lessons

Given a student opens a lesson

When the "Play Audio" button is clicked

Then the lesson should be read aloud.

US4 – Screen Reader

Given the page is loaded

When a screen reader is enabled

Then all interface elements should be accessible.

US5 – Voice Navigation

Given voice recognition is active

When the student says "Open Quiz"

Then the quiz page should open.

US6 – Accessible Quiz

Given the student starts a quiz

When answers are submitted

Then the score should be displayed.

US7 – Progress

Given quizzes are completed

When the dashboard opens

Then the latest progress should be shown.

US8 – Manage Users

Given an administrator logs in

When selecting Manage Users

Then user information should be displayed and editable.

US9 – Upload Content

Given the administrator uploads a PDF or audio file

When upload completes

Then students should be able to access it.

US10 – Reports

Given quiz data exists

When the administrator generates a report

Then student performance should be displayed.

4. Product Backlog

Priority	User Story	Business Value
High	Student Registration	High
High	Login	High
High	Audio Lessons	High
High	Screen Reader Support	High
High	Voice Navigation	High
Medium	Accessible Quiz	Medium
Medium	Progress Tracking	Medium
Medium	Manage Users	Medium
Low	Upload Learning Materials	Low
Low	Generate Reports	Low

5. Sprint Backlog (Sprint 1)

Sprint Duration

2 Weeks

Selected User Stories

- US1 Registration
- US2 Login
- US3 Audio Lessons
- US4 Screen Reader Support
- US5 Voice Navigation

Development Tasks

Task	Assigned Feature
Design Login Page	Login
Develop Registration Module	Registration
Create Dashboard	Dashboard
Implement Text-to-Speech	Audio Lessons
Integrate Voice Navigation	Voice Commands
Test Accessibility	Screen Reader

6. Story Point Estimation (Fibonacci)

User Story	Story Points	Reason
Registration	3	Simple form and validation
Login	3	Authentication logic
Audio Lessons	5	Browser speech integration
Screen Reader Support	5	Accessibility testing required
Voice Navigation	8	Speech recognition implementation
Quiz	5	Questions and scoring
Progress Tracking	5	Database operations
Manage Users	8	CRUD operations
Upload Content	5	File handling

Reports	8	Analytics and reporting
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7. Sprint Goal

Develop the core features required for visually impaired students to access learning materials independently.

8. Sprint Deliverables

- Student Registration
- Secure Login
- Student Dashboard
- Audio Learning Module
- Voice Navigation
- Screen Reader Support
- SQLite Database
- Flask Backend
- HTML/CSS User Interface
- Initial Accessibility Testing

9. Definition of Done

A user story is considered complete when:

- Development is completed.
- Code is reviewed.
- Unit testing is successful.
- Accessibility requirements are verified.
- User interface is responsive.
- Documentation is updated.
- Product Owner approves the feature.
- No critical defects remain.

10. Conclusion

The **Accessible Learning Platform** demonstrates how Agile software engineering practices can be applied to develop an inclusive educational system. By using sprint planning, user stories, product backlogs, and iterative development, the project delivers essential accessibility features such as screen reader support, voice navigation, audio learning resources, and accessible assessments. This solution promotes equal learning opportunities for students with visual impairments and supports **SDG 4 – Quality Education** by making digital education more accessible, inclusive, and user-friendly.